Ojas Varshney

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.79
GVN The Global School	Class XII (CBSE)	2023	94.80%
GVN The Global School	Class X (CBSE)	2021	92.00%

WORK EXPERIENCE

Tower Research Capital

Core Engineering Intern -- Market Data Core

- Integrated offline trade-to-trade (T2T) analytics directly into the core ATOM production engine, eliminating cross-codebase duplication through high-speed parsers that process full data streams without message loss.
- Designed a **protocol-agnostic** market data parser using advanced **C++26 Reflection** to automatically identify, structure, and parse arbitrary exchange messages at compile time.
- Built a **generic C++26 reflection-based serialize**r, successfully eliminating **10-20 thousand** lines of hardcoded boilerplate while enabling zero-overhead automated schema updates.
- Expanded inline stream-capture capabilities to enable unified serialization and T2T delay profiling on unfiltered feeds, fully maintaining hot-path low latency.

PROJECTS

LOKUS Quant DSL

- Built a quantitative trading DSL compiler translating **SMA**, **RSI**, **ATR**, and regime logic into **MLIR.**
- Implemented a multi-phase compiler using Flex, Bison/Yacc, C++ ASTs, symbol tables, and semantic type checking.
- Designed hybrid DSL semantics combining immutable declarative trading rules with a C-like imperative core for numerical tasks.

Multicast and Wireless Network-on-Chip in gem5

- Extended gem5's Ruby memory system with in-network multicast support, optimizing data distribution across a NoC.
- Built a configurable synthetic multicast traffic generator emitting packets with randomly sized destination sets, plumbed end-to-end through the SLICC coherence layer into the network interface.
- Extending the fabric with wireless support where a single broadcast transmission serves an entire multicast group.

RISC-V Environment

- Developed a complete RISC-V environment in C++ including an assembler, simulator, and configurable cache manager.
- Implemented assembly-level instruction support with label handling, error checking, and standard RISC-V ISA compatibility, facilitating accurate program translation.
- Designed a flexible cache simulator with tunable parameters for exploring cache performance and memory management.

High-Performance Multi-Threaded Quiz Server

- Engineered a concurrent C++ quiz server utilizing socket programming, POSIX threads (pthreads), and mutex lock synchronization to manage state, answer validation, and dynamic leaderboard generation across concurrent clients.
- Implemented an LFU caching layer with concurrent "in-generation" state locks, eliminating duplicate LLM API calls and boosting throughput by **over 160%** under high concurrent load (10 clients).
- Designed a custom low-overhead TCP protocol with prefix-based parsing and non-blocking asynchronous worker threads.

Vehicle Rental System

- Developed an advanced SQL-based system enabling end-to-end online management of vehicle rentals.
- Implemented real-time vehicle availability, reservation, booking management, and transaction features.
- Integrated comprehensive admin tools for inventory tracking, customer records management, dynamic pricing adjustments, and real-time business metrics, streamlining operational workflows.

SKILLS

- **Languages & Scripting:** C, C++ (with OOP), Shell Scripting (Bash), MySQL, Verilog, RISC-V, HTML, CSS, Markdown
- **Development & Build Environments:** Linux, Visual Studio Code, Vim, CMake, Buck2
- **Version Control & Tooling:** Git/GitHub, Gerrit, GDB, LaTeX

NOTABLE ACHIEVEMENTS

- **Codeforces Expert:** Achieved title of Expert with peak rating of 1658 on Codeforces platform.
- **NTSE Scholar:** Recipient of prestigious NTSE Scholarship awarded by NCERT, Govt. of India.
- **IMO International Rank 1:** Secured International Rank 1 in International Mathematics Olympiad by SOF.
- **RMO Qualified:** Qualified for Regional Mathematics Olympiad (RMO).

RELEVANT COURSES

Introduction to Programming*, Probability and Random Variables, Discrete Mathematics for Computer Science, Theory of Computation, Introduction to Statistics, Data Structures*, Algorithms, Software Development Fundamentals, Operating Systems, Database Management Systems, Computer Architecture*, Computer Networks, Compilers, Foundations of Machine Learning*, Computational Number Theory and Linear Algebra, Linear Optimisation. (*with a lab component.)

POSITIONS OF RESPONSIBILITY

- Served as a Teaching Assistant for the course Introduction to Programming (ID1063).
- Selected for the post of Badminton Coordinator at IIT Hyderabad.

EXTRACURRICULAR ACHIEVEMENTS

- FIDE Rated chess player with ratings of 1456 (Classical), 1498 (Blitz).
- Represented International Board of School Organisation (IBSO) for Badminton in SGFI Nationals.